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INTRODUCTION

RETIREMENT PLANNERS:

- * "As investors age, they should start cutting back on riskier investments" (Malkiel, 1999)

MODERN PORTFOLIO THEORY:

- * "in your prime of life you have the same relative risk-tolerance as toward the end of life!" (Samuelson, 1969)
- * Constant strategy, independent of t and T . (Merton, 1969)

$$dS_t^\circ = S_t^\circ (\mu dt + \sigma dB_t)$$

TURNPIKE:

- * Despite joint dependence on the horizon and market states there is a fixed limit for large horizons - typically.

MODEL - DYNAMICS

THE MARKET:

- * safe asset earning zero interest rate
- * risky asset with price S_t

$$dS_t = \mu dt + dX_t,$$

$$dX_t = -\alpha \operatorname{sgn}(X_t) |X_t|^\beta dt + dB_t,$$

$$S_0 = X_0 = 0, \beta \geq 1, \alpha > 0, \mu \in \mathbb{R}, B_t$$

MODEL - COMMENTS

- * The case $\beta = 1$ recovers a trending O-U process.
- * Nonlinear mean reversion:

$$d(S_t - \mu t) = -\text{sgn}(S_t - \mu t)|S_t - \mu t|^\beta dt + dB_t$$

- * The case $\alpha = 0$ recovers the Bachelier model, Black-Scholes
- * The case $\alpha > 0$, fading price shocks, X_t has stationary density

$$\frac{e^{-\frac{\alpha}{\beta+1}|x|^{\beta+1}}}{2\left(\frac{\alpha}{\beta+1}\right)^{-\frac{1}{1+\beta}}\Gamma\left(\frac{2+\beta}{1+\beta}\right)}$$

- * Two different regimes

PRELIMINARIES

* PROBLEM:

$$E \left[-e^{-(H \cdot S)_T} \right] \rightarrow \max$$

* VALUE FUNCTION:

$$u_T = \sup_{H \in \mathcal{X}_T} E \left[-e^{-(H \cdot S)_T} \right]$$

* CERTAINTY EQUIVALENT:

$$c_T = -\log(-u_T)$$

* ORDER OF CERTAINTY EQUIVALENT:

$$\mathcal{C}(\mathfrak{H}) = \sup \left\{ \theta : \liminf_{T \rightarrow \infty} \frac{-\log(E[e^{-(H(T) \cdot S)_T}])}{T^\theta} > 0 \right\}$$

HEURISTICS - A DISCRETE-TIME RESULT

AR(1):

$$* \nu_0 \in (-1, 1), \quad R_k = \nu_0 R_{k-1} + \eta_k.$$

$$* \text{ Set } \nu = \nu_0 - 1, \quad R_k - R_{k-1} = \nu R_{k-1} + \eta_k, \quad k \geq 1,$$

PROBLEM:

$$* (H \cdot R)_N = \sum_{j=1}^N H_j (R_j - R_{j-1}), \quad U(x) = -\exp\{-x\},$$

$$u_N = \sup_H E \left[-e^{-(H \cdot R)_N} \right]$$

* Looking for H^* such that

$$E \left[-e^{-(H^* \cdot R)_N} \right] = u_N$$

* How the certainty equivalent grows:

$$c_N = -\ln(-u_N)$$

HEURISTICS - A DISCRETE-TIME RESULT

THEOREM (Rásonyi, Deák, 2015) :

- * The optimal investment for time horizon N is

$$H_{k,N}^* := g_{k,N}(R_{k-1}), \quad 1 \leq k \leq T,$$

where

$$g_{k,N}(z) = \nu z [1 - (N - k)\nu] \quad \text{for all } 1 \leq k \leq N \text{ and } z \in \mathbb{R}.$$

- * These strategies, with $\nu \neq 0$, yield the maximal expected utilities

$$\begin{aligned} u_N = u(N; \nu) &= \sup_{H \in \mathcal{X}_N} \mathbb{E} \left[-e^{-(H \cdot R)_N} \right] \\ &= \mathbb{E} \left[-e^{-(H^* \cdot R)_N} \right] = -\bar{\gamma}(N; \nu)^{-\frac{1}{2}}, \end{aligned}$$

where $\bar{\gamma}(N; \nu) := \nu^{2N} \Gamma(1/\nu^2 + N) \Gamma(1/\nu^2)^{-1}$.

HEURISTICS - COMMENTS

- * For $\nu \neq 0$: $c_N = -\ln(-u_N)$ grows as $N \ln(N)$.
- * Investments with homogeneous strategies can only have a growth of order N .
- * Investments with temporal structure can have a larger growth rate.
- * In a martingale model there is no growth.
- * Mean-reversion is exploited in the beginning.

HEURISTICS - ORNSTEIN-UHLENBECK PROCESS

$$dS_t = -\alpha S_t dt + dB_t.$$

- * Euler approximation on $[0, T]$ with resolution N :

$$\bar{R}_k^{(N)} = \bar{R}_{k-1}^{(N)} - \alpha \frac{T}{N} \bar{R}_{k-1}^{(N)} + \bar{\eta}_k,$$

- * Transform to match parameters:

$$\bar{R}_k^{(N)} - \bar{R}_{k-1}^{(N)} = -\alpha \frac{T}{N} \bar{R}_{k-1}^{(N)} + \bar{\eta}_k$$

Growth rate of the approximation with resolution N :

$$T \mapsto u(N; -\frac{\alpha T}{N})$$

HEURISTICS - ORNSTEIN-UHLENBECK PROCESS

Taking the limit $N \rightarrow \infty$

$$\left(T \mapsto u\left(N; -\frac{\alpha T}{N}\right) \right) \rightarrow \left(T \mapsto -e^{-\alpha^2 T^2/4} \right).$$

COMMENTS:

- * The limit accidentally exists.
- * The "what is the growth rate" and the "limiting operation" is not necessarily interchangeable.
- * How robust is Euler's method.

CLOSED FORM SOLUTION - O-U PROCESS

ORNSTEIN-UHLENBECK WITH DRIFT:

$$dS_t = \mu dt + dX_t$$

$$dX_t = -\alpha X_t dt + \sigma dB_t$$

PROBLEM:

$$* (H \cdot S)_T = \int_0^T H_t dS_t,$$

$$u_T = \sup_H E \left[\frac{e^{-\gamma(H \cdot S)_T}}{-\gamma} \right]$$

* Looking for H^* such that

$$E \left[-e^{-(H^* \cdot R)_T} \right] = u_T$$

* Certainty equivalent:

$$c_T = \frac{\ln(-\gamma u_T)}{-\gamma}$$

CLOSED FORM SOLUTION - O-U PROCESS

$$* H_t = \frac{\mu - \alpha X_t}{\gamma \sigma^2} + \frac{\mu - \alpha X_t}{\gamma \sigma^2} \alpha (T - t) + \frac{\mu \alpha^2}{2 \gamma \sigma^2} (T - t)^2$$

$$* c_T = \frac{\mu^2}{2 \gamma \sigma^2} T + \left(\frac{\alpha \mu^2}{2 \gamma \sigma^2} + \frac{\alpha^2}{4 \gamma} \right) T^2 + \frac{\mu^2 \alpha^2}{6 \gamma \sigma^2} T^3$$

SETUP

ORDER OF c_T

$$\alpha \neq 0 \text{ and } \mu \neq 0, \quad T^3$$

$$\mu \neq 0 \text{ but } \alpha = 0, \quad T^1$$

$$\alpha = 0 \text{ and } \mu = 0 \quad T^0$$

THE CASE OF THE HEURISTIC LIMIT: If $\mu = 0$ but $\alpha \neq 0$ then

$$c_T = \frac{\alpha^2}{4} T^2$$

THEOREM 1 - HOW TO TRADE FORD SHARES:

$$dS_t = \mu dt + dX_t,$$

$$dX_t = -\alpha \operatorname{sgn}(X_t) |X_t|^\beta dt + dB_t,$$

- * If $\mu \neq 0$, then

$$c_T \leq C_{\beta, \mu, \alpha} T^{2\beta+1}$$

for some constant $C_{\beta, \mu, \alpha} > 0$, and each family of strategies $\mathfrak{H}$ satisfies

$$\mathcal{C}(\mathfrak{H}) \leq 2\beta + 1.$$

- * The family of strategies $\mathfrak{H}_\beta = (H(\beta, T))_{T>0}$ defined by

$$H_t(\beta, T) = (\beta + 1)(T - t)^\beta \operatorname{sgn}(S_t) |S_t|^\beta, \quad t \in [0, T]$$

satisfies

$$\mathcal{C}(\mathfrak{H}_\beta) = 2\beta + 1.$$

THEOREM 2 - HOW TO TRADE GOLD:

- * If $\mu = 0$, then

$$c_T \leq C_{\beta,\alpha} T^{\beta+1}$$

for some constant $C_{\beta,\alpha} > 0$, and each family of strategies $\mathfrak{H}$ satisfies

$$\mathcal{C}(\mathfrak{H}) \leq \beta + 1.$$

- * If $\beta > 1$, then for each $1 < \gamma < \beta$, the family of strategies $\mathfrak{H}_\gamma = (H(\gamma, T))_{T>0}$ defined by

$$H_t(\gamma, T) = -2(T - t)^\gamma S_t, \quad t \in [0, T]$$

satisfies

$$\mathcal{C}(\mathfrak{H}) \geq 1 + \gamma.$$

- * If $\beta = 1$ then

$$\mathcal{C}(\mathfrak{H}_1) = 2 = 1 + \beta.$$

COMMENTS:

- * The strategies grow without bounds, no turnpike
- * No fixed annuity is sufficient
- * A myopic strategy, calibration at the start
- * Insights from the Ornstein-Uhlenbeck case
- * Same decline in risk-taking, but smaller order

PROOF:

PROPOSITION. The process

$$\xi_t^* = e^{-\int_0^t (\mu - \alpha \operatorname{sgn}(X_u) |X_u|^\beta) dB_u - \frac{1}{2} \int_0^t (\mu - \alpha \operatorname{sgn}(X_u) |X_u|^\beta)^2 du}$$

is a P -martingale and $dQ_T/dP = \xi_T^*$ defines a probability measure $Q_T \sim P$ on $\mathcal{F}_T$ such that $(S_t)_{t \in [0, T]}$ is a Q_T -martingale and Q_T is the only such probability measure.

PROOF. By Girsanov's theorem we only have to establish that ξ^* is a true martingale. This can be done utilizing the result (Mijatović and Urusov, 2012).

PROOF:

DUALITY THEORY OF OPTIMAL INVESTMENT:

(Delbaen et al., 2002, Kabanov and Stricker, 2002)

$$J = E \left[\frac{dQ_T}{P} \log \left(\frac{dQ_T}{P} \right) \right] = E[\xi_T^* \log(\xi_T^*)] = E_{Q_T}[\log(\xi_T^*)]$$

FORMULA FOR THE VALUE FUNCTION:

$$u_T = -e^{-J}$$

PROOF ($\mu = 0$):

DYNAMICS WITHOUT DRIFT:

$$dS_t = dX_t = -\alpha \operatorname{sgn}(X_t) |X_t|^\beta dt + dB_t$$

THE DENSITY PROCESS:

$$\begin{aligned} \log(\xi_T^*) &= \int_0^T \alpha \operatorname{sgn}(X_u) |X_u|^\beta dB_u - \frac{1}{2} \int_0^T \alpha^2 |X_u|^{2\beta} du \\ &= \frac{\alpha^2}{2} \int_0^T |X_u|^{2\beta} du + \alpha \int_0^T \operatorname{sgn}(X_u) |X_u|^\beta dX_u \end{aligned}$$

- * The process X is a standard Brownian motion on $[0, T]$ under the measure Q_T .

$$J = \frac{\alpha^2}{2} \int_0^T E_{Q_T}[|X_u|^{2\beta}] du = \frac{\alpha^2}{2(1+\beta)} T^{1+\beta}$$

PROOF:

ESTIMATES:

(Kontoyiannis et al., 2005)

DEFINITION:

$$\mathcal{A}f = -\alpha \operatorname{sgn}|x|^\beta \partial_x f + \frac{1}{2} \partial_{xx} f. \quad f \in C^2(\mathbb{R})$$

$$\mathcal{H}f = e^{-f} \mathcal{A}e^f, \quad f \in C^2(\mathbb{R})$$

CONDITION:

There is a compact $C \subset \mathbb{R}$, there are $\delta, b > 0$ and functions $V, W : \mathbb{R} \mapsto \mathbb{R}_+$ with W measurable and $V \in C^2(\mathbb{R})$ such that, for all $x \in \mathbb{R}$,

$$\mathcal{H}V(x) \leq -\delta W(x) + b \mathbb{1}_C(x).$$

PROOF:

DEFINITION:

$$M_t = \exp \left\{ V(X_t) + \int_0^t (\delta W(X_u) - b \mathbb{1}_{X_u \in C}) du \right\}, \quad t \in \mathbb{R}_+$$

LEMMA:

If the **CONDITION** holds then the process M is a supermartingale.

PROOF.

Defining $Z_t = \exp \left\{ \int_0^t (\delta W(X_u) - b \mathbb{1}_{X_u \in C}) du \right\}$ we have

$$\begin{aligned} dM_t &= \\ &= Z_t e^{V(X_t)} (\delta W(X_t) - b \mathbb{1}_{X_t \in C} + \mathcal{H}V(X_t)) dt + Z_t (e^V \partial_x V)(X_t) dB_t \end{aligned}$$

PROOF:

COROLLARY:

With the **CONDITION** in force for $T > 0$ we have

$$E \left[\exp \left\{ \int_0^T \delta W(X_u) du \right\} \right] \leq e^{V(0)+bT}.$$

DEFINITION:

$$\bar{V}(x) = \frac{\alpha}{1+\beta} |x|^{1+\beta} \quad \text{and} \quad \bar{W}(x) = \alpha^2 |x|^{2\beta}, \quad x \in \mathbb{R}$$

PROPOSITION:

For each $0 < \bar{\delta} < 1/2$, there is an appropriate constant $\bar{b} > 0$ and a compact set $\bar{C}$ such that the **CONDITION** is fulfilled with $V = \bar{V}$, $W = \bar{W}$, $b = \bar{b}$, $\delta = \bar{\delta}$ and $C = \bar{C}$.

PROOF:

LEMMA: There exists constants $\delta_0, c_0, C_0 > 0$ such that

$$E \left[\exp \left\{ \delta_0 \int_0^T |X_t|^{2\beta} dt \right\} \right] \leq c_0 e^{C_0 T}.$$

PROOF ($\mu = 0, \beta = 1$):

Considering the process $\bar{U}_t = \delta_0(T - t)X_t^2$, with differentiation we have

$$0 = U_T - U_0 = \int_0^T -\delta_0 X_t^2 dt + \int_0^T 2\delta_0(T - t)X_t dX_t + \int_0^T \delta_0(T - t)dt,$$

reordering yields

$$\int_0^T -2\delta_0(T - t)X_t dX_t = -\int_0^T \delta_0 X_t^2 dt + \int_0^T \delta_0(T - t)dt.$$

This is the value of the portfolio utilizing the strategy

$$\bar{H}_t = -2\delta_0(T - t)X_t.$$

with portfolio value

$$E \left[-e^{-(H \cdot S)_T} \right] = -e^{-\frac{\delta_0}{2} T^2} E \left[e^{\int_0^T \delta_0 X_t^2 dt} \right]$$

that is, with **Lemma**, we have

$$E \left[-e^{-(H \cdot S)_T} \right] \geq -e^{-\frac{\delta_0}{2} T^2 + C_0 T} \quad \blacksquare$$

PROOF ($\mu = 0, \beta > 1$):

For $1 < \gamma < \beta$, consider the process $U_t = (T - t)^\gamma X_t^2$, $t \in [0, T]$.

Since $U_0 = 0$ we have

$$0 = U_T = \int_0^T 2(T-t)^\gamma X_t dX_t + \int_0^T (T-t)^\gamma dt - \int_0^T \gamma(T-t)^{\gamma-1} X_t^2 dt$$

which is equivalent to

$$\int_0^T -2(T-t)^\gamma X_t dX_t = \frac{1}{\gamma+1} T^{\gamma+1} - \int_0^T \gamma(T-t)^{\gamma-1} X_t^2 dt.$$

This is the value of the portfolio utilizing the strategy

$$H_t(\gamma, T) = -2(T-t)^\gamma X_t,$$

hence

$$E \left[-e^{-(H \cdot S)_T} \right] = -e^{-\frac{1}{\gamma+1} T^{\gamma+1}} E \left[\exp \left\{ \int_0^T \gamma(T-t)^{\gamma-1} X_t^2 dt \right\} \right],$$